

ALI HAIDER

Simulation Software Engineer

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📍 Gulistan e Jauhar., Karachi

ABOUT ME

Simulation Software Engineer specializing in Unity, interactive 3D environments, Digital Twins, and cross-platform mobile development. Experienced in designing custom toolchains, C#/.NET architecture, database integrations, and mobile solutions using React Native. Combines strong technical skills with instructional experience, focusing on scalable software design and workflow efficiency across enterprise and mobile platforms.

EDUCATION

MAJU - Virtual University | 2022-2026

BS-Software Engineering

Transferred academic credits from Mohammad Ali Jinnah University (MAJU) to Virtual University of Pakistan to complete Bachelor of Science in Software Engineering coursework. Focused on core software engineering principles, algorithms, database management systems, and object-oriented programming.

Aptech Education | 2015-2018

ACCP PRO - 3 Years Diploma

Completed a comprehensive 3-year ACCP Pro career program specializing in professional software engineering, modern development frameworks, web applications, mobile applications, and database management systems. Gained hands-on expertise in software architecture, system design, and database administration.

WORK EXPERIENCE

The Location Lab | 2023 – Present

Simulation Software Engineer

Engineered real-time 3D simulations, Digital Twins, and VR environments using Unity. Built custom Editor toolchains to optimize development pipelines and integrated DBMS backends for scalable performance.

Aptech Computer Education | 2021 – 2023

Teaching Faculty

Taught software engineering, web application development, and database systems. Mentored students through practical coding projects and software design principles.

NAVTTTC | 2022 – 2022

Game Development Instructor

Taught 3D game mechanics and Unity development, guiding trainees through hands-on project creation and industry practices.

SKILLS

- Game Programming
- Level Designing
- VR | ML Agents | Dots
- Digital Twins
- Unity Editor Tools
- 3d Simulations
- .Net C#
- React Native
- DBMS

PROJECTS

Enterprise 3D Industrial Simulation Platform

Developed a real-time 3D simulation suite combining an interactive railway training simulator with dynamic fault injection, a terrain-adaptive power pylon and wire tension planning tool, and an interactive machinery disassembly engine for technical inspection and component swapping to drastically reduce field risks, planning costs, and physical survey overhead.

Automated Industrial Logistics & 4D Digital Twin

Engineered a real-time 3D simulation for scheduled chemical and material logistics, automated queuing, and gate operations. Built a custom time-scrubbing playback controller that dynamically aligns environmental daylight and recalculates all active vehicle states, positions, and unloading tasks forward or backward to any arbitrary timestamp.

High-Fidelity Environment & Level Design

Designed and built high-fidelity 3D environments ranging from atmospheric horror landscapes to coastal and mountainous terrains using Unity HDRP. Configured physical lighting systems, volumetric fog, post-processing stacks, and terrain compositions to create immersive visual storytelling while maintaining smooth rendering performance.

Box Buddy-Mobile 2.5D Arcade Game (Available on Playstore)

Architected and published a mobile arcade game on Google Play to master the end-to-end Android deployment and store release lifecycle. Built multiple dynamic game modes in an optimized hybrid 2D/3D environment, integrating responsive physics-based movement, thematic level variants, and Unity Ads monetization.

Interactive Gameplay Prototypes & Systems

Developed Unity C# gameplay prototypes featuring procedural maze generation algorithms, a low-poly 3D mobile adventure with custom character animations and fluid movement mechanics, and modular puzzle game frameworks (Sudoku, Word Search, Wordly) engineered with scalable architectures for multiplayer expansion.

2D Concept Art & Level Design Implementation

signed atmospheric 2D concept art and modular environment assets in Adobe Photoshop, structuring layered cityscapes and platforming elements. Implemented the complete 2D level pipeline into Unity, integrating multi-plane parallax backgrounds, dynamic lighting, and suspended platform mechanics to create a cohesive visual style and smooth gameplay flow.

EDITOR TOOLS - UNITY

- Hierarchy Management Tool
- Physics based rope creator
- Vegetation system
- File Reader (XML,Docx,Xls,Text,Json)

RUNTIME TOOLS - UNITY

- Sandbox
- Measurement Tool
- Traffic System With Signals
- Weather System